

Operational Amplifier (Op-Amp).

- ✳ **Op-Amp:** Is a high gain differential amplifier with a high input impedance and low output impedance.
- ✳ **Why it is called operational amplifier (Op – Amp)?**

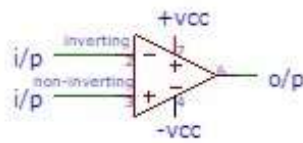
An operational amplifier, or op-amp, is called that because it was originally designed to perform mathematical operations like addition, subtraction, integration, and differentiation in analog computers. Today, it's widely used in various electronic circuits due to its ability to amplify signals and perform these operations efficiently.

- ✳ **Typical uses:**

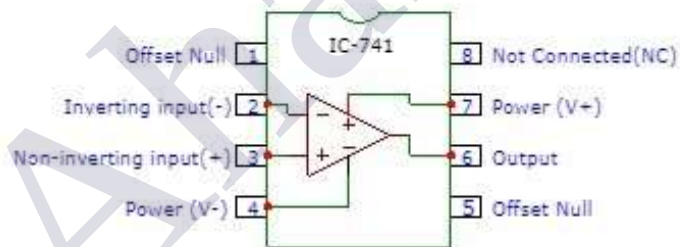
- ✓ To provide voltage amplitude changes.
- ✓ Oscillators.
- ✓ Filter circuits.
- ✓ Many types of instrumentation circuits.

- ✳ **Show the Circuit symbol and pin configuration of a typical op-amp with its function.**

The symbol and pin configuration of op-amp 741 is shown below. The diagram comprises eight pins. Among these, pins 2,3, and 6 are the most significant pins, where pins 2 and 3 represent inverting and non-inverting terminals respectively, and pin 6 represents output voltage. Pin 8 is inactive in the circuit. The number 741 in the name indicates that there are 7 active pins, 4 pins (pin 2,3,4,7) are capable of taking input and 1 pin (pin 6) is an output pin. The triangular form in the IC represents an op-amp integrated circuit.



Op-Amp IC 741 Symbol



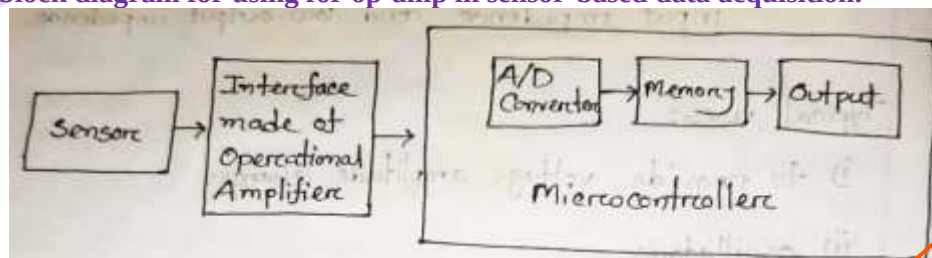
Op-Amp IC 741 Pin Diagram

How2Electronics

Functionality of the pin diagram:

- **Pin 1 and 5 (Offset Null):** Used to adjust the offset voltage.
- **Pin 2 (Inverting Input):** Receives the input signal to be inverted.
- **Pin 3 (Non-Inverting Input):** Receives the input signal to be amplified without inversion.
- **Pin 4 (V- or Negative Power Supply):** Connects to the negative power supply.
- **Pin 6 (Output):** Gives the amplified output signal.
- **Pin 7 (V+ or Positive Power Supply):** Connects to the positive power supply.
- **Pin 8 (NC - Not Connected):** Not used in the circuit.

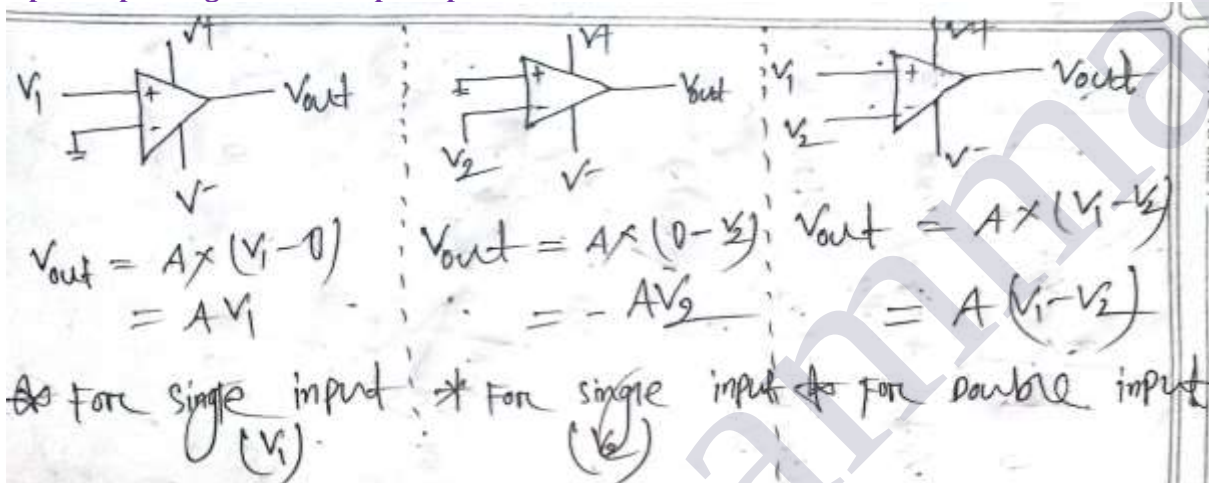
- ✳ **Show the block diagram for using for op-amp in sensor-based data acquisition.**



★ Op-Amp Characteristics.

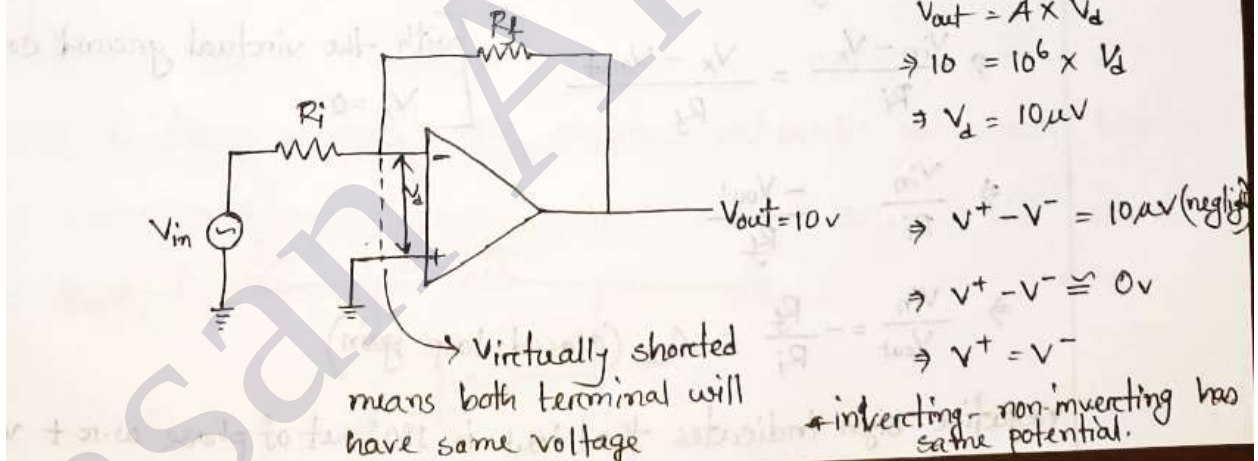
- ✓ Infinite input impedance. ($R_i = \infty$)
- ✓ Output impedance. ($R_o = 0$)
- ✓ Infinite open-loop gain. ($A = \infty$)
- ✓ $V_{out} = 0$ when $V_{in} = 0$
- ✓ Infinite bandwidth.
- ✓ Infinite Common mode rejection rate (CMRR).

★ Open loop configuration of Op-Amp.



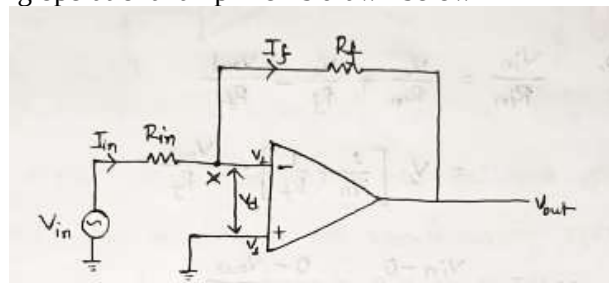
★ Inverting Op-Amp configuration.

Inverting Op-amp Configuration



★ Draw the circuit for an inverting amplifier. Deduce the equation of output voltage, current and closed loop gain for inverting op-amp.

The circuit of the inverting operational amplifier is drawn below-

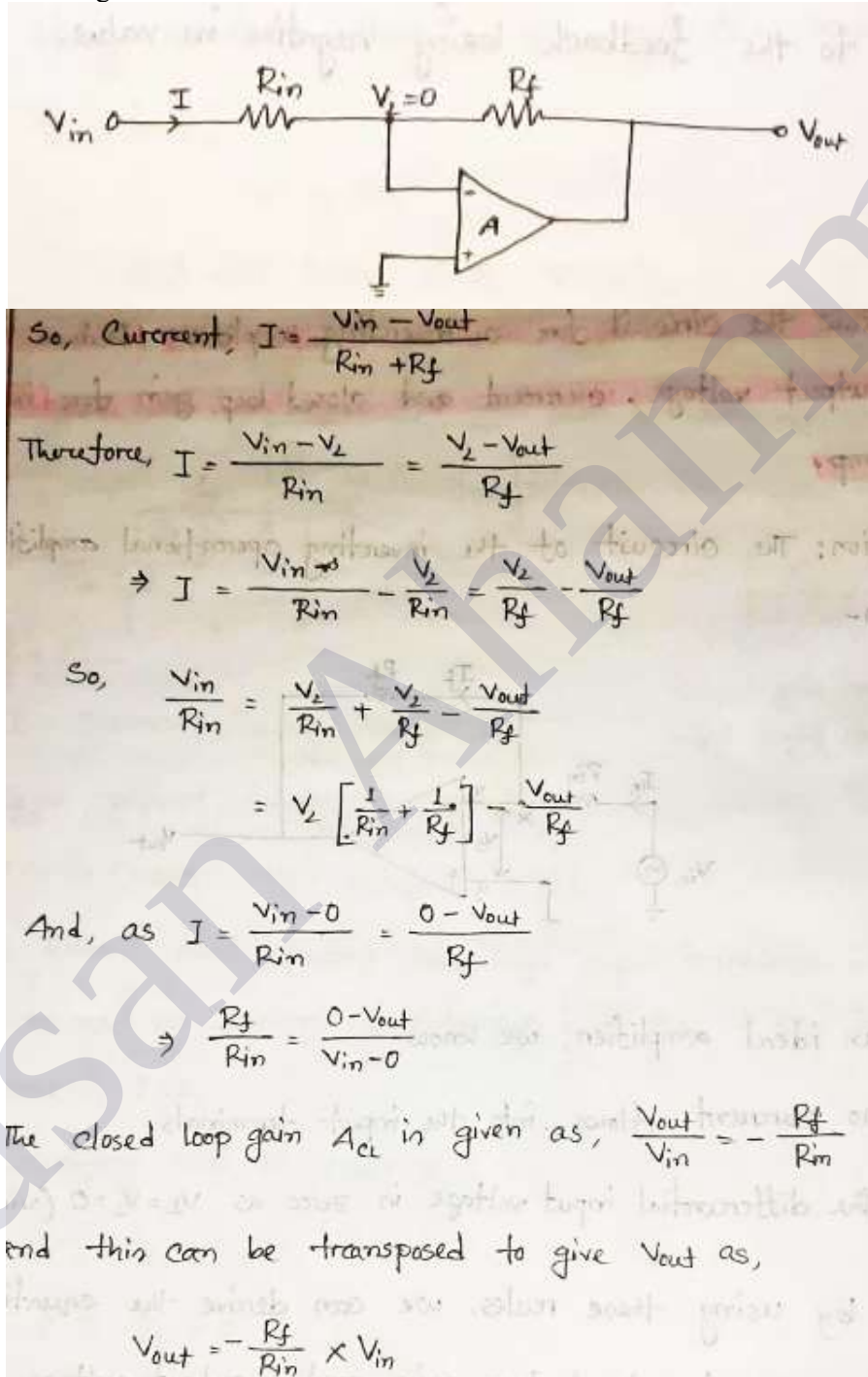


For an ideal amplifier, we know

- i) No current flows into the input terminals.
- ii) The differential input voltage is zero as $V_1 = V_2 = 0$ (virtual ground)

Then, by using these rules, we can drive the equations for calculating current, closed loop gain and output voltage of inverting amplifier.

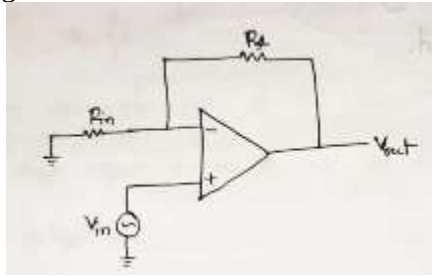
Current 'I' flows through the resistor network as shown below-



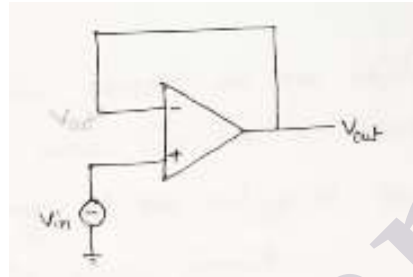
The negative sign is the indication of the inversion of the output signal w.r.t input signal as it is 180° out of phase. It is due to the feedback being negative in value.

✳ Explain how an operational amplifier can be used to design voltage follower?

The non-inverting op-amp configuration is,



This operational amplifier can be used to design a voltage follower. If we make the $R_f=0$ and $R_{in}=\infty$ then the circuit becomes like this:



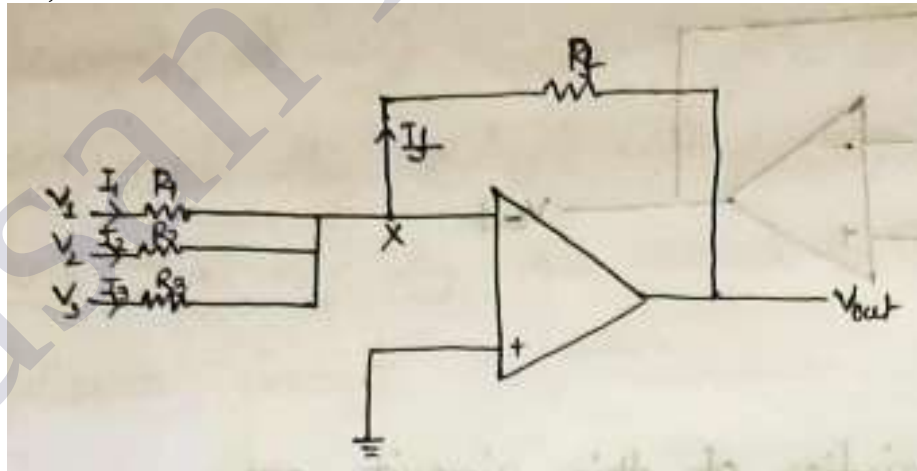
Here, we have shorted this output terminal with the inverting terminal. So, at the inverting terminal the voltage become V_{out} . Because of the feedback, we have virtual short between inverting and non-inverting terminal. So we can say that,

$$V^- = V^+ \\ V^- = V_{in} = V_{out}$$

It means, whatever voltage is applied to the non-inverting terminal, it will appear at the output voltage. So, we can say that, the output voltage follows the input voltage and that is why this circuit is called as the voltage follower circuit.

✳ How an Op-Amp can be used to build inverting adder?

Inverting adder,



We know, whenever there is a negative feedback to the op-amp then there is a virtual short between inverting and non-inverting terminal. So, if one terminal is at ground potential then another terminal also will be at the ground potential. Hence, the node x will have 0 potential.

Now applying KCL at node 'x',

$$I_1 + I_2 + I_3 = I_f$$

$$\Rightarrow \frac{V_1 - 0}{R_1} + \frac{V_2 - 0}{R_2} + \frac{V_3 - 0}{R_3} = \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} = - \frac{V_{out}}{R_f}$$

$$\Rightarrow V_{out} = - \left[\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2} V_2 + \frac{R_f}{R_3} V_3 \right]$$

Now let us assume $R_1 = R_2 = R_3 = R$, then

$$V_{out} = - \frac{R_f}{R} [V_1 + V_2 + V_3]$$

And if $R_f = R$, then

$$V_{out} = - [V_1 + V_2 + V_3]$$

In this way we can use op-amp as adder and we can add multiple input voltages.

★ **Inverting adder uses:**

- ✓ Summing, averaging, scaling.
- ✓ Providing DC offset.
- ✓ Digital to analog converter (DAC)
- ✓ Audio mixer.

★ **Draw the circuit for a non-inverting amplifier and explain its operations.**

An amplifier which amplifies the input without producing any phase shift between input and output is called **Noninverting Amplifier**.

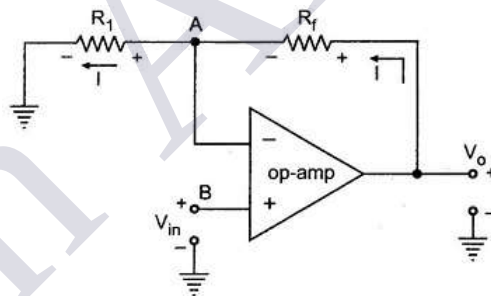


Fig. 2.20 Noninverting amplifier

The positive sign indicates that there is no phase shift between input and output. The input and output waveforms are shown in the Fig. 2.21.

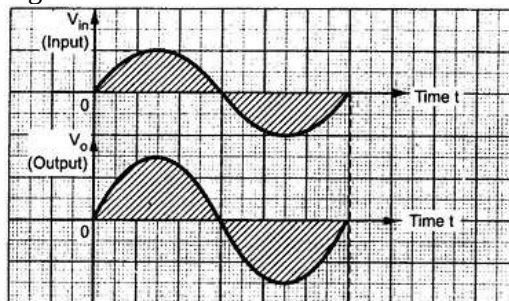
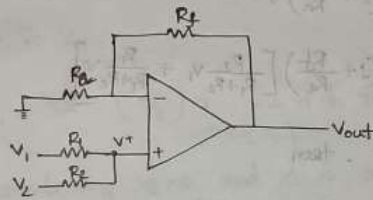


Fig. 2.21

✧ Non-inverting adder.



For a non-inverting configuration, we know

$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) \times V^+$$

As we have applied multiple inputs at V^+ , so we have to calculate V^+ by superposition theorem,

First let us assume, V_1 is acting alone and $V_2=0$. So in that case the voltage at the node is V_1^+ , so,

$$V_1^+ = \frac{R_2}{R_1 + R_2} \times V_1$$

Likewise when we consider V_2 , and $V_1=0$, then

$$V_2^+ = \frac{R_1}{R_1 + R_2} \times V_2$$

So, $V^+ = V_1^+ + V_2^+$

$$= \frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2$$

Now,

$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) V^+$$

$$= \left(1 + \frac{R_f}{R_a}\right) \left[\frac{R_2}{R_1 + R_2} V_1 + \frac{R_1}{R_1 + R_2} V_2 \right]$$

Let, $R_1 = R_2 = R$, then

$$V_{out} = \left(1 + \frac{R_f}{R_a}\right) \left(\frac{V_1 + V_2}{2} \right)$$

Again, letting $R_f = R_a$, then

$$V_{out} = V_1 + V_2$$

which is the addition of the input voltage V_1 and V_2 .

✧ Subtractor operational amplifier.

Subtractor Op-amp



This circuit is the combination of inverting and non-inverting configuration. Applying superposition theorem for V_{out} . Let us consider $V_2 = 0$ and V_1 is acting alone. In this configuration, the output voltage is V_{o1} .

$$V_{o1} = \left(-\frac{R_2}{R_1}\right) V_1 \quad [\text{as it is the inverting configuration}]$$

Similarly, considering $V_1 = 0$ and V_2 is acting alone, then the output voltage of this configuration V_{o2} is,

$$V_{o2} = \left(1 + \frac{R_2}{R_1}\right) \times V_2$$

So, if these are acting simultaneously, then

$$\begin{aligned} V_{out} &= V_{o1} + V_{o2} \\ &= \left(-\frac{R_2}{R_1}\right) \times V_1 + \left(1 + \frac{R_2}{R_1}\right) \times V_2 \end{aligned}$$

Here we can see that, the gain for both cases are different. If we want the gain to be same for both inputs, and if we want the V_{out} as $K(V_2 - V_1)$ i.e. the difference between two input voltages times K , then we need to make sure the gain of the two input voltages are equal. So, in order to make the term $\left(1 + \frac{R_2}{R_1}\right)$ equal to $\frac{R_2}{R_1}$.



$$\begin{aligned} 1 + \frac{R_2}{R_1} &= \frac{R_1 + R_2}{R_1} \\ &= \frac{R_1 + R_2}{R_1} \times \frac{R_2}{R_1 + R_2} = \frac{R_2}{R_1} \end{aligned}$$

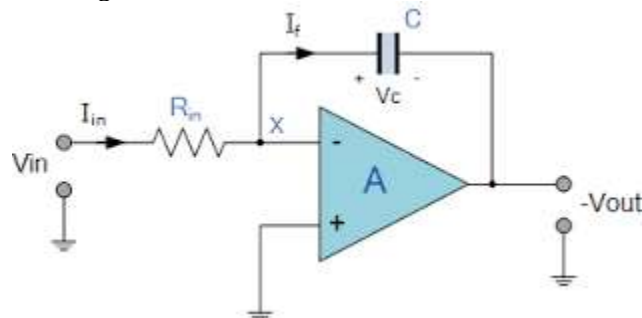
Here, we have applied the voltage divider instead of applying the voltage source V_2 directly. So, Now

$$\begin{aligned} V_{out} &= \left(-\frac{R_2}{R_1}\right) \times V_1 + \left(\frac{R_2}{R_1}\right) \times V_2 \\ \Rightarrow V_{out} &= \frac{R_2}{R_1} (V_2 - V_1) \end{aligned}$$

So, here the output voltage is opera the difference between the two voltage multiplied by the factor of $\frac{R_2}{R_1}$.

✳ Show the circuit diagram of integrator and differentiator and explain how it works.

- **Op-amp Integrator:** if we replace the feedback resistor of the inverting configuration with a capacitor, then we get the integrator circuit.



Let us consider the current flowing through R is I_{in} and current flowing through the capacitor is $I_f = I_c$. The capacitor C_F is connected as the feedback. Because of the feedback, the node x is virtually grounded. So, at this node the potential is 0. Applying KCL to the node,

$$I_{in} - I_c = 0$$

$$\Rightarrow I_{in} = I_c$$

$$\Rightarrow \frac{V_{in} - 0}{R} = I_c$$

We know capacitor current can be given by,

$$I_c = C \frac{dV_c}{dt}$$

where, V_c is the voltage at capacitor charge across the capacitor is,

Putting this value,

$$\begin{aligned} \frac{V_{in}}{R} &= C_F \times \frac{dV_c}{dt} \\ \Rightarrow \frac{V_{in}}{R} &= C_F \times \frac{d}{dt} [0 - V_{out}] \\ &= -C_F \times \frac{dV_{out}}{dt} \end{aligned}$$

$$\begin{aligned} Q &= CV \\ \Rightarrow \frac{dQ}{dt} &= C \frac{dV_c}{dt} = I_c \quad \text{[taking derivative]} \\ &\text{rate of change of charge} \end{aligned}$$

$$\frac{dV_{out}}{dt} = -\frac{1}{RC_F} \times V_{in}(t)$$

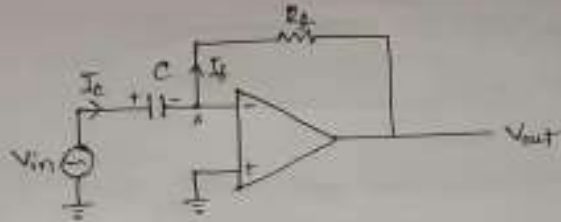
Taking integration for both sides,

$$V_{out}(t) = -\frac{1}{RC_F} \int V_{in}(t) dt$$

Here by this equation, we get the integration of input voltage times the factor $\frac{1}{RC_F}$ as the output voltage.

➤ **Op-amp differentiator:**

If we interchange the position of the capacitor and resistor of the integrator circuit, we get a simple differentiator circuit.



The current flowing through the capacitor is I_c and through R_f is I_f . We assume that this is an ideal op-amp, so no current is flowing in this op-amp. Due to the negative feedback, the node x is virtually grounded and its potential is 0.

Applying KCL at x ,

$$I_c - I_f = 0$$

$$\Rightarrow I_c = I_f$$

$$\Rightarrow C \frac{dV_x}{dt} = I_f \frac{0 - V_{out}}{R_f}$$

$$\Rightarrow C \frac{dV_{in}}{dt} = - \frac{V_{out}}{R_f}$$

$$\Rightarrow V_{out} = -R_f \times C \times \frac{dV_{in}}{dt}$$

So, in this circuit, the output voltage is the differentiation of the input voltage times $R_f C$ factor.

☆ How voltage gain is obtained for inverting amplifier?

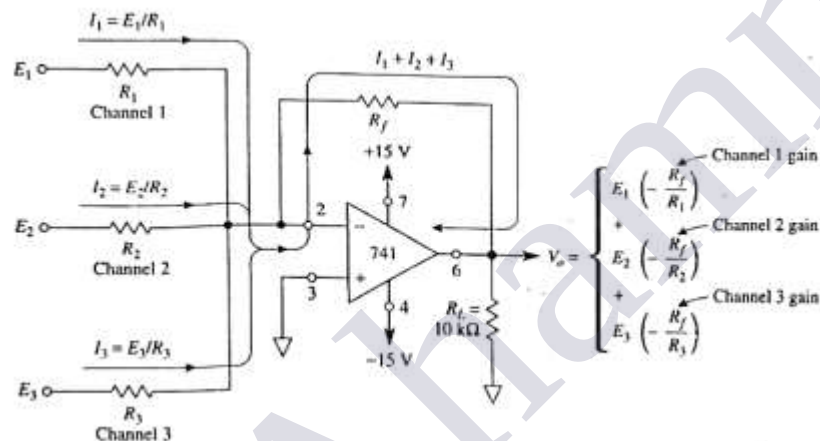
In an **inverting amplifier**, the voltage gain is determined by the ratio of two resistors:

$$\text{Gain} = -R_f/R_{in}$$

- **R_f** is the resistor connected between the output and the inverting input.
- **R_{in}** is the resistor connected to the input signal.

The **negative sign** means the output is flipped or "inverted" compared to the input.

☆ Draw the circuit diagram of a multichannel amplifier.



☆ Show how a low pass filter is designed using the operational amplifier?

An **active low-pass filter** allows **low frequencies** to pass through and **blocks high frequencies** using an **operational amplifier (op-amp)**, a **resistor (R)**, and a **capacitor (C)**.

- **Capacitor (C):** Passes low-frequency signals and blocks high frequencies.
- **Resistor (R):** Controls the filtering effect, working with the capacitor to set the **cutoff frequency** (the point where high frequencies start being blocked).
- **Op-amp:** Amplifies the output signal, providing better performance and gain compared to a passive filter.

Active Low Pass Filter

